

♥ HUMAN REPRODUCTION ♥

Reproduction is a biological process by which living organisms produce new individuals of their own kind.

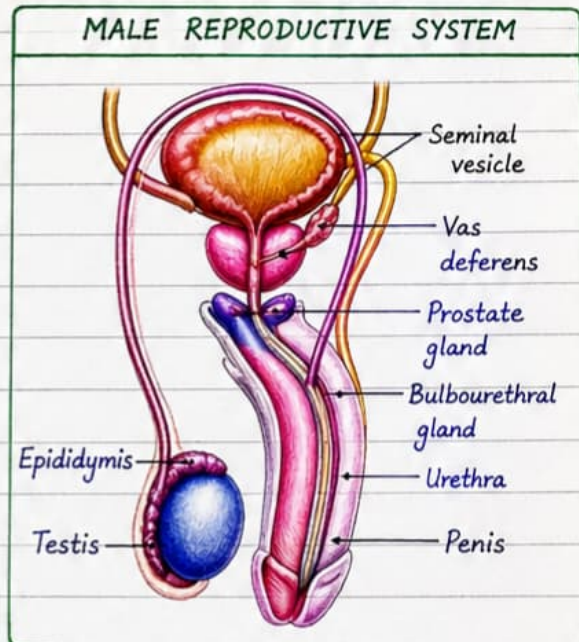
1. INTRODUCTION

- ★ Humans are sexually reproducing organisms.
- ★ Reproduction ensures **continuity** of species.
- ★ It involves production of **gametes**, their fusion and development of new individual.
- ★ It is a cyclic & regulated process controlled by **hormones**.

2. MALE REPRODUCTIVE SYSTEM

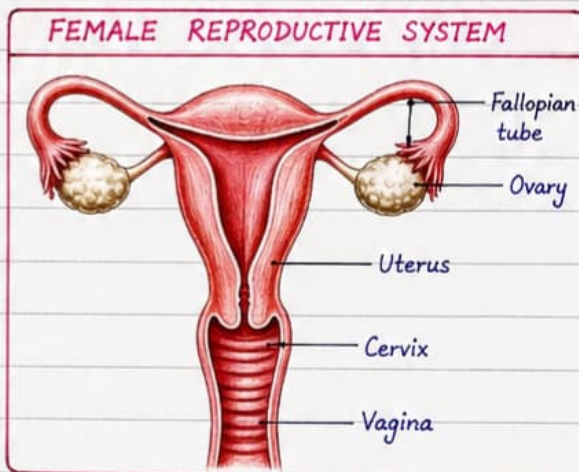
It consists of pair of **testes**, accessory ducts, glands and external genitalia.

- Testes** - Produce **sperm** & **testosterone**.
- Epididymis** - Stores & matures **sperm**.
- Vas deferens** - Transports **sperm**.
- Seminal vesicles** - Secrete **fructose-rich** fluid.
- Prostate gland** - Adds milky **alkaline** fluid.
- Bulbourethral glands** - Lubricates urethra.
- Urethra** - Common passage for urine & semen.
- Penis** - Copulatory organ.

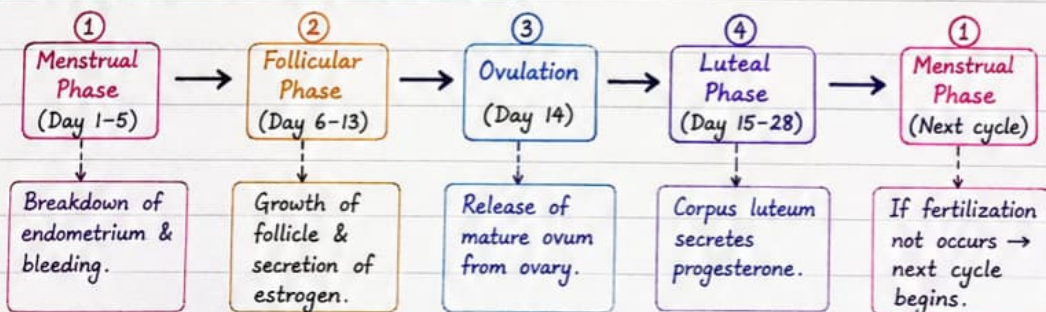


3. FEMALE REPRODUCTIVE SYSTEM

- Ovaries** - Produce **ova** & hormones (**estrogen**, **progesterone**).
- Fallopian tubes** - Pick up ovum, site of **fertilization**.
- Uterus** - Site of **implantation** & development of **embryo**.
- Cervix** - Lower narrow part of uterus.
- Vagina** - Copulatory organ & birth canal.
- Vulva** - External genitalia.



4. REPRODUCTIVE CYCLE IN FEMALE



KEY POINTS

- ★ **Fertilization** → Fusion of sperm & ovum.
- ★ **Implantation** → Blastocyst gets embedded in uterus.
- ★ **Gestation period** in humans → ~280 days.

Reproduction is not just a process, it is the continuity of life. ♥

GAMETES, FERTILIZATION & IMPLANTATION

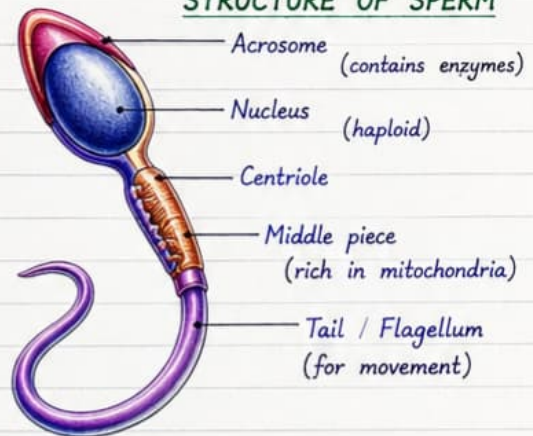
1. SPERM (MALE GAMETE)

- ★ Produced in seminiferous tubules of testes.
- ★ Each sperm is microscopic, motile & streamlined.
- ★ Structure → Head, Middle piece, Tail
- ★ Head contains nucleus & **acrosome** (enzymes help in fertilization).
- ★ Middle piece has many mitochondria (provide energy).
- ★ Tail (flagellum) helps in movement.

NOTE

Millions of sperms are released, but only one sperm can fertilize one ovum.

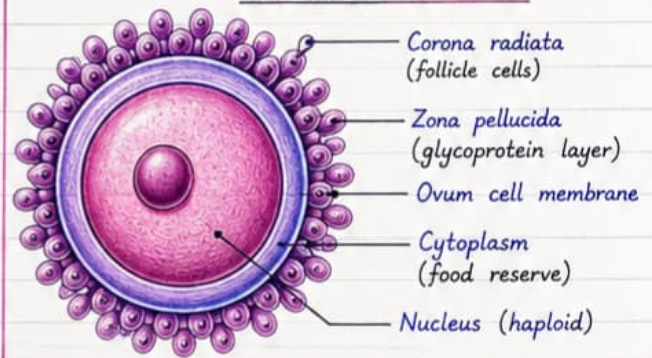
STRUCTURE OF SPERM



2. OVUM (FEMALE GAMETE)

- ★ Produced in ovaries.
- ★ Largest cell in human body.
- ★ Non-motile & spherical.
- ★ Surrounded by protective layers.
- ★ Contains rich cytoplasm (food reserve) for early development.

STRUCTURE OF OVUM



COMPARISON

	Sperm	Ovum
Size	Very small	Very large
Number	Millions	One per cycle
Motility	Motile	Non-motile
Cytoplasm	Almost absent	Abundant
Function	Fertilizes ovum	Nourishes embryo.

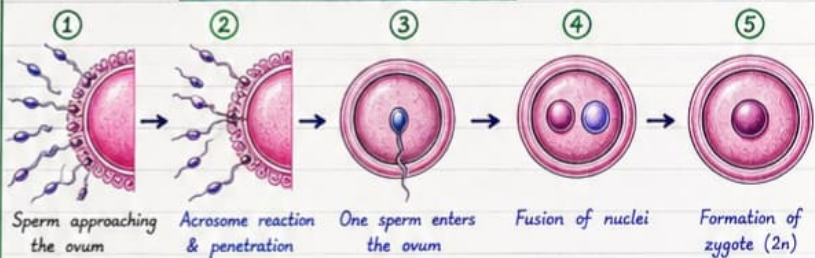
3. FERTILIZATION

- ★ Fusion of male gamete (sperm) with female gamete (ovum).
- ★ Takes place in **ampullary part** of fallopian tube.
- ★ Only one sperm enters the ovum.
- ★ Results in formation of **zygote (2n)**.
- ★ Rest of sperms degenerate.

STEPS :

- ① Sperm reaches near ovum (chemotaxis).
- ② **Acrosome reaction** → enzymes help sperm to penetrate layers.
- ③ Fusion of sperm membrane with ovum membrane.
- ④ Cytoplasm & nuclei fuse → **ZYGOTE**.

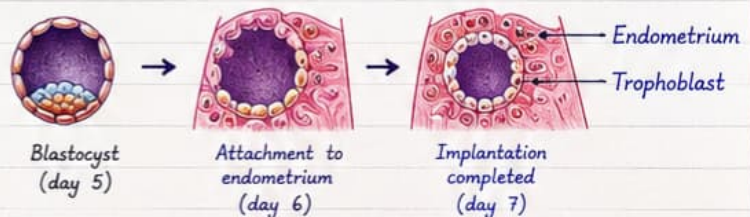
STEPS OF FERTILIZATION



IMPORTANT

Fertilization restores diploid number & determines the genetic sex of the baby (X from ovum, X or Y from sperm).

IMPLANTATION OF BLASTOCYST



4. IMPLANTATION

- ★ The zygote divides repeatedly to form **blastocyst** (day 5-6).
- ★ Blastocyst reaches uterus & gets implanted in thickened endometrium (day 6-7).
- ★ Trophoblast helps in implantation & forms placenta later.
- ★ Inner cell mass develops into embryo.

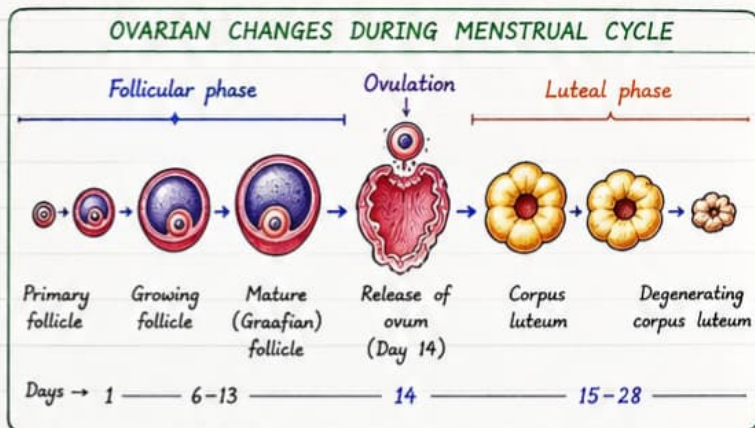
SUMMARY

Gametes (n) → Fertilization (in oviduct) → Zygote (2n) → Divisions → Blastocyst → Implantation (in uterus)

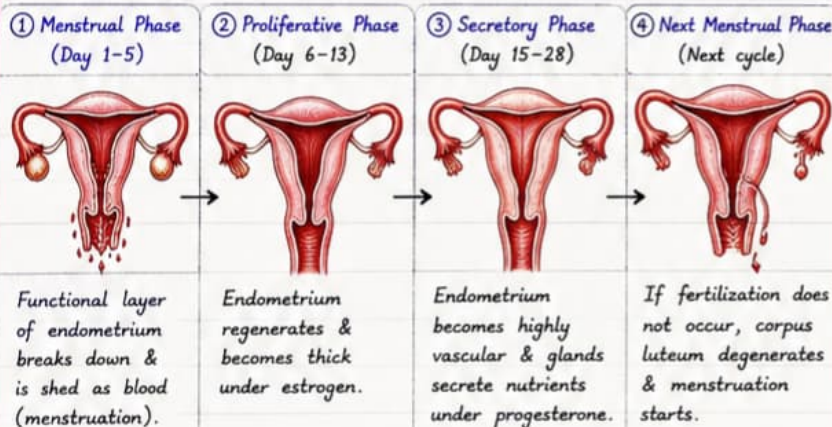
♥ MENSTRUAL CYCLE & HORMONAL CONTROL ♥

1. MENSTRUAL CYCLE (Average 28 days)

- ★ Cyclic changes in ovary and uterus.
- ★ Controlled by hormones from pituitary gland & ovaries.
- ★ Four main phases -
 - ① Menstrual phase (Day 1-5)
 - ② Follicular / Proliferative phase (Day 6-13)
 - ③ Ovulatory phase (Day 14)
 - ④ Luteal / Secretory phase (Day 15-28)



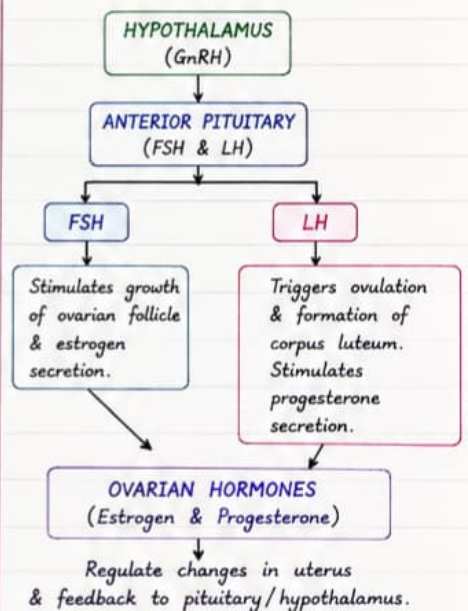
2. UTERINE CHANGES



4. HORMONES & THEIR FUNCTIONS

HORMONE	SOURCE	FUNCTION
GnRH	Hypothalamus	Stimulates pituitary to release FSH & LH.
FSH	Anterior pituitary	Growth of ovarian follicle & secretion of estrogen.
LH	Anterior pituitary	Causes ovulation & formation of corpus luteum; secretion of progesterone.
Estrogen	Ovarian follicle	Proliferation of endometrium; development of female secondary sexual characters.
Progesterone	Corpus luteum	Maintains endometrium; prepares uterus for implantation.

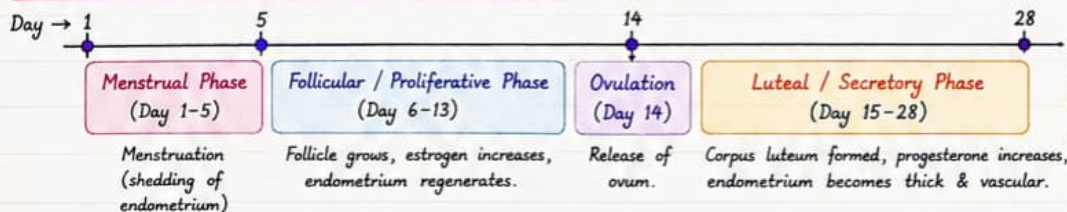
3. HORMONAL CONTROL



IMPORTANT POINTS

- ★ Ovulation occurs on Day 14.
- ★ Fertilization usually occurs in fallopian tube.
- ★ If fertilization does not occur, corpus luteum degenerates & menstruation begins.
- ★ The cycle repeats every ~28 days.

5. MENSTRUAL CYCLE TIMELINE (OVERVIEW)



If no fertilization → corpus luteum degenerates → hormone levels fall → menstruation starts. New cycle begins!

QUICK REVISION

- ✓ Sperms are produced in testes; ovum in ovaries.
- ✓ Fertilization → Zygote (2n) → Implantation → Embryo.
- ✓ Ovarian & uterine changes are controlled by hormones.
- ✓ Menstrual cycle is a cyclic hormonal phenomenon (~28 days).

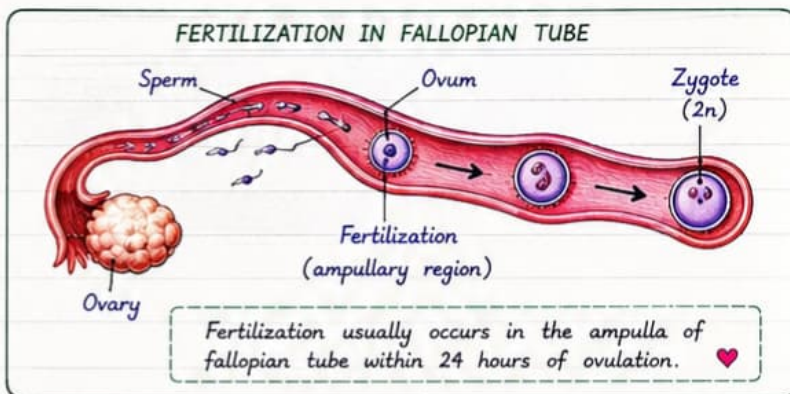
★ REMEMBER ★

"Reproduction is the key to continuity, variation is the key to evolution."

FERTILIZATION TO EMBRYO DEVELOPMENT

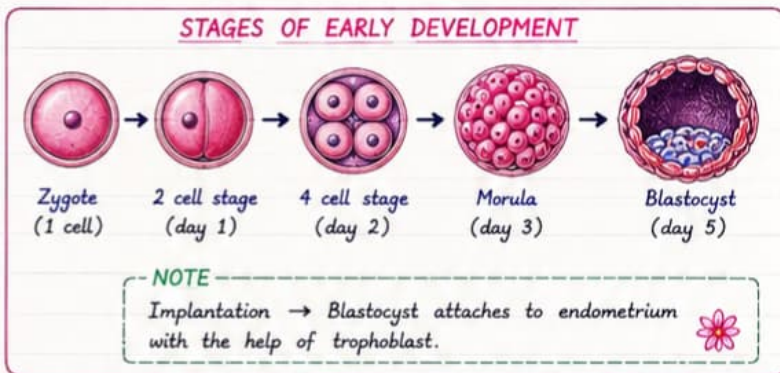
1. JOURNEY OF SPERM & FERTILIZATION

- ★ During coitus, millions of sperms are deposited in **vagina**.
- ★ They move through cervix → uterus → fallopian tube (**ampullary region**).
- ★ Only one sperm penetrates the ovum.
- ★ The sperm & ovum membranes fuse.
- ★ Male & female pronuclei fuse → **Zygote (2n)**.



2. CLEAVAGE & EMBRYO DEVELOPMENT

- ★ The zygote undergoes rapid mitotic divisions called **cleavage** while moving towards uterus.
- ★ Size remains same but number of cells increases.
- ★ Forms a solid ball of cells called **morula**.
- ★ Morula transforms into **blastocyst**.
- ★ Blastocyst reaches uterus in 4-5 days.
- ★ Implantation of blastocyst occurs in the endometrium of uterus (day 6-7).

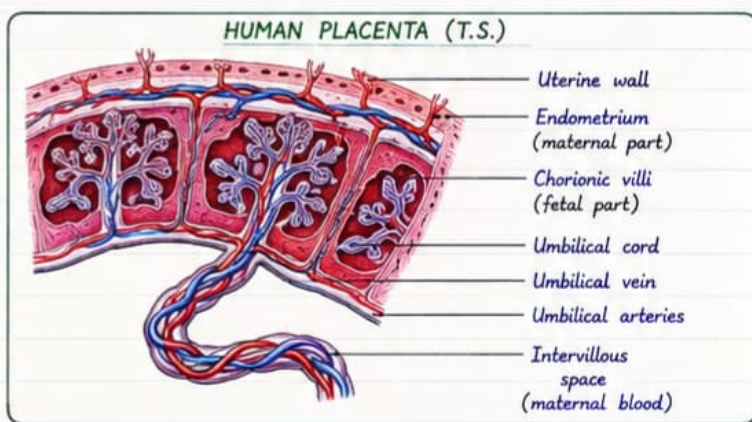


3. IMPLANTATION & PLACENTA FORMATION

- ★ The trophoblast of blastocyst invades the endometrium.
- ★ Inner cell mass (**embryoblast**) gives rise to embryo.
- ★ Trophoblast develops chorionic villi which form the fetal part of placenta.
- ★ Endometrium of mother forms maternal part of placenta.
- ★ Placenta connects foetus with uterine wall and is linked by the umbilical cord.

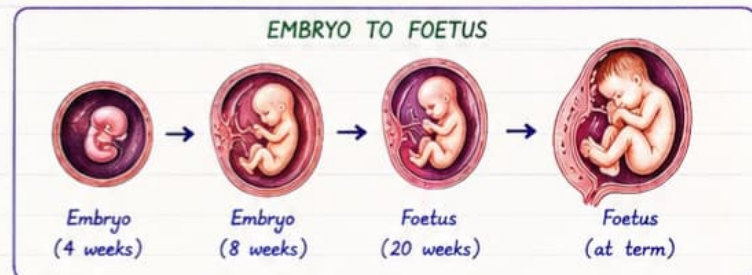
IMPORTANCE OF PLACENTA

- ★ Provides nutrients & oxygen to foetus
- ★ Removes carbon dioxide & metabolic wastes.
- ★ Secretes hormones - hCG, progesterone, estrogen, hPL.
- ★ Acts as barrier & protects from many infections.
- ★ Transfers antibodies from mother → foetus.



4. EMBRYO DEVELOPMENT (WEEKS)

- ★ **Week 1-2** → Zygote → Cleavage → Implantation.
- ★ **Week 3-8** → Organogenesis (formation of organs).
- ★ **Week 9 onwards** → Foetus stage (growth & maturation).
- ★ By **38-40 weeks** → Full term baby is ready for birth.



DIFFERENCE BETWEEN EMBRYO & FOETUS

Feature	Embryo (up to 8 weeks)	Foetus (after 8 weeks)
Structure	Organogenesis stage	Organs already formed
Appearance	Small, like a cartilaginous structure	Human appearance visible
Growth	Rapid cell division	Increase in size & weight
Functions	Formation of organs	Maturation of organs
Movement	Not visible	Movements start

QUICK FACTS

- ★ Life begins at fertilization.
- ★ Zygote is totipotent.
- ★ Implantation occurs on day 6-7.
- ★ Menstrual cycle average ~28 days.
- ★ Full term pregnancy ~280 days. (9 months 10 days)

A tiny beginning of life creates a miracle called human.

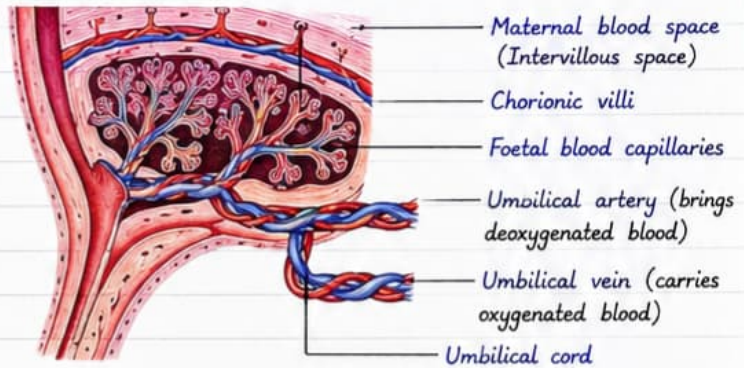
★ Reproduction is the continuity of life; development makes it beautiful.

♥ PREGNANCY, PARTURITION & LACTATION ♥

1. PREGNANCY

- ★ Implantation of blastocyst in endometrium marks the beginning of pregnancy.
- ★ Embryo gets embedded in uterine wall (day 6-7).
- ★ Chorionic villi develop & form placenta.
- ★ Placenta acts as an interface between mother & foetus.
- ★ It provides nutrients, O_2 & removes waste.
- ★ It also secretes hormones hCG, estrogen & progesterone to maintain pregnancy.

HUMAN PLACENTA (STRUCTURE & FUNCTIONS)



Placenta is temporary but life-saving organ for the developing baby. ♥

2. HORMONES DURING PREGNANCY

HORMONE	SOURCE	FUNCTION
hCG	Trophoblast cells of placenta	Maintains corpus luteum till placenta takes over.
Progesterone	Corpus luteum & later placenta	Maintains endometrium, prevents contractions.
Estrogen	Placenta	Growth of uterus, breasts & blood supply.
hPL (Human placental lactogen)	Placenta	Prepares mammary glands for lactation; alters mother's metabolism.

3. PARTURITION (CHILDBIRTH)

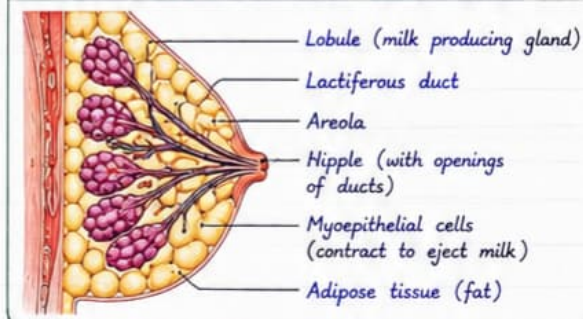
- ★ It is the process of giving birth to baby.
- ★ Hormone oxytocin (from posterior pituitary) causes strong rhythmic contraction of uterus.
- ★ Positive feedback - More contraction → More oxytocin release.
- ★ Cervix dilates, baby moves through birth canal & is delivered.
- ★ After delivery of baby, placenta is expelled out (called afterbirth).



4. LACTATION (MILK PRODUCTION)

- ★ Milk is produced in mammary glands of both breasts.
- ★ Hormone prolactin (from anterior pituitary) stimulates milk production.
- ★ Oxytocin causes contraction of myoepithelial cells → milk ejection (let-down reflex).
- ★ Breastfeeding provides ideal nutrition, immunity & emotional bonding.

STRUCTURE OF MAMMARY GLAND (T.S.)



REFLEX ARC

Suckling of baby (stimulus)
↓
Nerve impulses from nipple
↓
Hypothalamus
↓
Posterior pituitary releases oxytocin
↓
Contraction of myoepithelial cells
↓
Milk ejection

5. SIGNIFICANCE OF REPRODUCTION

- ✓ Ensures continuity of species.
- ✓ Maintains genetic variation.
- ✓ Plants & animals produce new individuals.
- ✓ Provides opportunity for evolution.
- ✓ Reproduction is essential for survival of life.

NEET QUICK POINTS

- ★ Fertilization is internal in humans.
- ★ Placenta connects foetus with uterine wall.
- ★ Gestation period in humans ≈ 280 days.
- ★ Oxytocin → uterine contraction & milk ejection.
- ★ Prolactin → milk production.

♥ Life is beautiful because it is continuous. ♥

